

10TH SCIENCE SURA GUIDE

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TAMIL NADU STATE BOARD - CLASS 10 SCIENCE
SURA GUIDE COMPLETE QUESTION BANK (PHYSICS, CHEMISTRY, BIOLOGY) & EVALUATION SCHEME

UNIT 1 (PHYSICS): Laws of Motion

1 Mark Objective Questions

Q1: The inertia of a body depends on...

Ans: Mass of the object (Inertia is directly proportional to mass). [1 Mark]

Q2: Impulse is equals to...

Ans: Change in linear momentum ($J = \Delta p$). [1 Mark]

2 Marks Short Answer Questions

Q3: State Newton's second law of motion.

Ans: The force acting on a body is directly proportional to the rate of change of linear momentum of the body and the change in momentum.

Q4: Differentiate between Mass and Weight.

Ans: 1) Mass: Fundamental quantity, scalar, measured in kg, remains constant everywhere. [1 Mark]

2) Weight: Derived quantity, vector, measured in Newton (N), varies with acceleration due to gravity ($W = mg$). [1 Mark]

7 Marks Comprehensive Board Question

Q5: State and prove the Law of Conservation of Linear Momentum.

Ans: Statement: In the absence of an external force, the total linear momentum of a system of bodies remains constant. [1 Mark]

Step 1: Setup: Let two bodies A and B of masses m_1 and m_2 move with initial velocities u_1 and u_2 ($u_1 > u_2$). [1 Mark]

Step 2: Force during collision: Body A exerts action force F_B on B, and B exerts reaction force F_A on A. [1 Mark]

Step 3: By Newton's Second Law: $F_B = m_2(v_2 - u_2)/t$ and $F_A = m_1(v_1 - u_1)/t$. [1 Mark]

Step 4: By Newton's Third Law: Action = -Reaction $\Rightarrow F_B = -F_A$. [1 Mark]

Step 5: Equating equations: $m_2(v_2 - u_2)/t = -m_1(v_1 - u_1)/t \Rightarrow m_2 v_2 - m_2 u_2 = -m_1 v_1 + m_1 u_1$. [1 Mark]

Step 6: Final Result: $m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$. Total initial momentum = Total final momentum. Hence proved. [1 Mark]

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TAMIL NADU STATE BOARD - CLASS 10 SCIENCE - Continued

UNIT 2 (PHYSICS): Optics & UNIT 7 (CHEMISTRY): Atoms and Molecules

1 Mark Questions

Q1: The refractive index of diamond with respect to air is approximately...

Ans: 2.42 (Diamond has very high refractive index). [1 Mark]

Q2: The value of Avogadro number is...

Ans: 6.023×10^{23} particles per mole. [1 Mark]

2 Marks Questions

Q3: State Snell's Law of Refraction.

Ans: The ratio of sine of angle of incidence to the sine of angle of refraction is equal to the ratio of refractive indices of the two media: $\sin i / \sin r = n_2 / n_1$

Q4: Calculate the number of moles in 27g of Aluminium (Atomic mass of Al = 27).

Ans: Number of moles = Mass / Atomic Mass = $27 / 27 = 1$ mole. [2 Marks]

4 Marks Long Answer Question

Q5: Differentiate between Myopia and Hypermetropia.

Ans: 1) Myopia (Short-sightedness): Nearby objects can be seen clearly; distant objects cannot. Eyeball lengthens. Focal length decreases.

2) Hypermetropia (Long-sightedness): Distant objects can be seen clearly; nearby objects cannot. Eyeball shortens. Focal length increases.